

Context: WWAALL is a multi-tenant B2B SaaS. Below are three short, fictionalized scenarios from our stack. For each, identify the problems and propose a fix written response, code welcome but not required.

Budget: ~2 hours total. If you only had time for one, prioritize

Scenario 1 — Backend / Domain (45-60 min)

```
namespace App\Domain\KeyResult;
use App\Models\KeyResult;
use App\Models\Workspace;
class KeyResultProgressUpdater
{
    public function updateProgress(int $keyResultId, float $newValue): KeyResult
    {
        $keyResult = KeyResult::find($keyResultId);
        if ($newValue < 0 || $newValue > $keyResult->target) {
            throw new \InvalidArgumentException('Invalid progress value');
        }
        $keyResult->current_value = $newValue;
        $keyResult->save();
        Cache::put("kr_progress_{$keyResultId}", $newValue, 3600);
        return $keyResult;
    }
}
```

Questions: What problems do you see? What's the concrete production risk of each? If you could only fix one before this shipped, which one, and why?

Scenario 2 — Infra / DevOps (30-40 min)

```
# bitbucket-pipelines.yml (excerpt)
deploy-production:
  - step:
      name: Deploy to ECS
      script:
        - export AWS_SECRET_ACCESS_KEY=AKIAxxxxxxxxxxxxxxxxxx
        - aws ecs update-service --cluster prod --service wwaall-api
        --force-new-deployment
        - echo "Deployed"
```

Questions: What would you flag here before approving this pipeline? What's missing to make this deploy safe to run unattended? What would you change first, and why?

Scenario 3 — AI-Assisted Engineering Workflow (20-30 min)

We run development through AI coding agents governed by a set of rules and hooks. Two engineers report a symptom: the same kind of commit gets blocked by an automated hook for one of them, but not the other and a project rule about branch naming contradicts a second rule in a different file, so agents apply one or the other unpredictably.

Questions: How would you diagnose this? What's the underlying failure mode (not the specific bug, the class of problem)? What would you change in how these rules/hooks are authored or governed to stop this from recurring?